

Group 1

Project presentation

Phytoplankton Air Systems

Algaerithms Inc.

Dhairya Pal (N01576099)
Julian Imperial (N01638310)
Dharmik Shah (N01581796)
Sanskriti Mansotra (N01523183)





Sanskriti M.
Scrum Master & Developer



Dhairya P.
Team Lead & Developer



Dharmik S.
Lead Developer & Database Manager



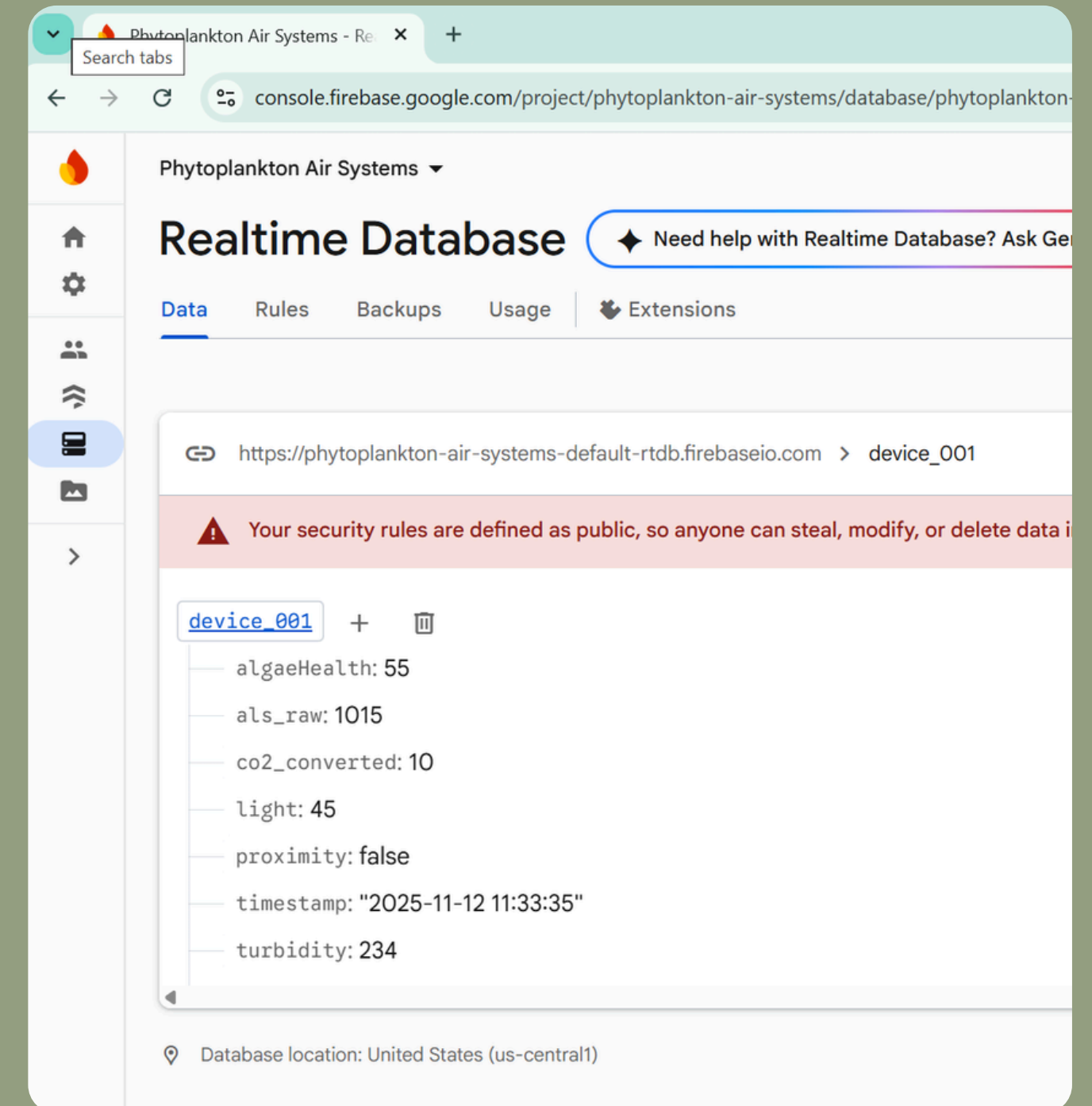
Julian I.
Developer & Agile Support

Hardware Architecture — Phytoplankton Air System

- Air Quality Sensor: Measures overall air quality levels inside the enclosure to support system monitoring.
- Light Sensor: Detects light intensity and environmental brightness for phytoplankton growth conditions.
- Proximity Sensor: Registers nearby motion or user interaction for system responses.
- Turbidity Sensor: Assesses water clarity and phytoplankton density in the chamber.
- Custom PCB + Raspberry Pi 4: Consolidates all sensor connections, reduces wiring errors, improves durability, and powers the entire system.

Cloud Database & Backend Structure

- The system uses Firebase to store and manage all sensor data and user information.
- Realtime Database holds live environmental readings—air quality, light, proximity, and turbidity—sent continuously from the Raspberry Pi.
- Data is uploaded using Firebase’s SDK for fast, synchronized updates with the mobile application.
- Firebase Authentication securely stores and manages user login credentials.
- All device data is organized under a dedicated data path, allowing the mobile app to retrieve and visualize values instantly.
- The backend supports multi-screen functionality including the Dashboard, Leaderboard, and Insights views.



Mobile Application — Real-Time Monitoring & User Interface

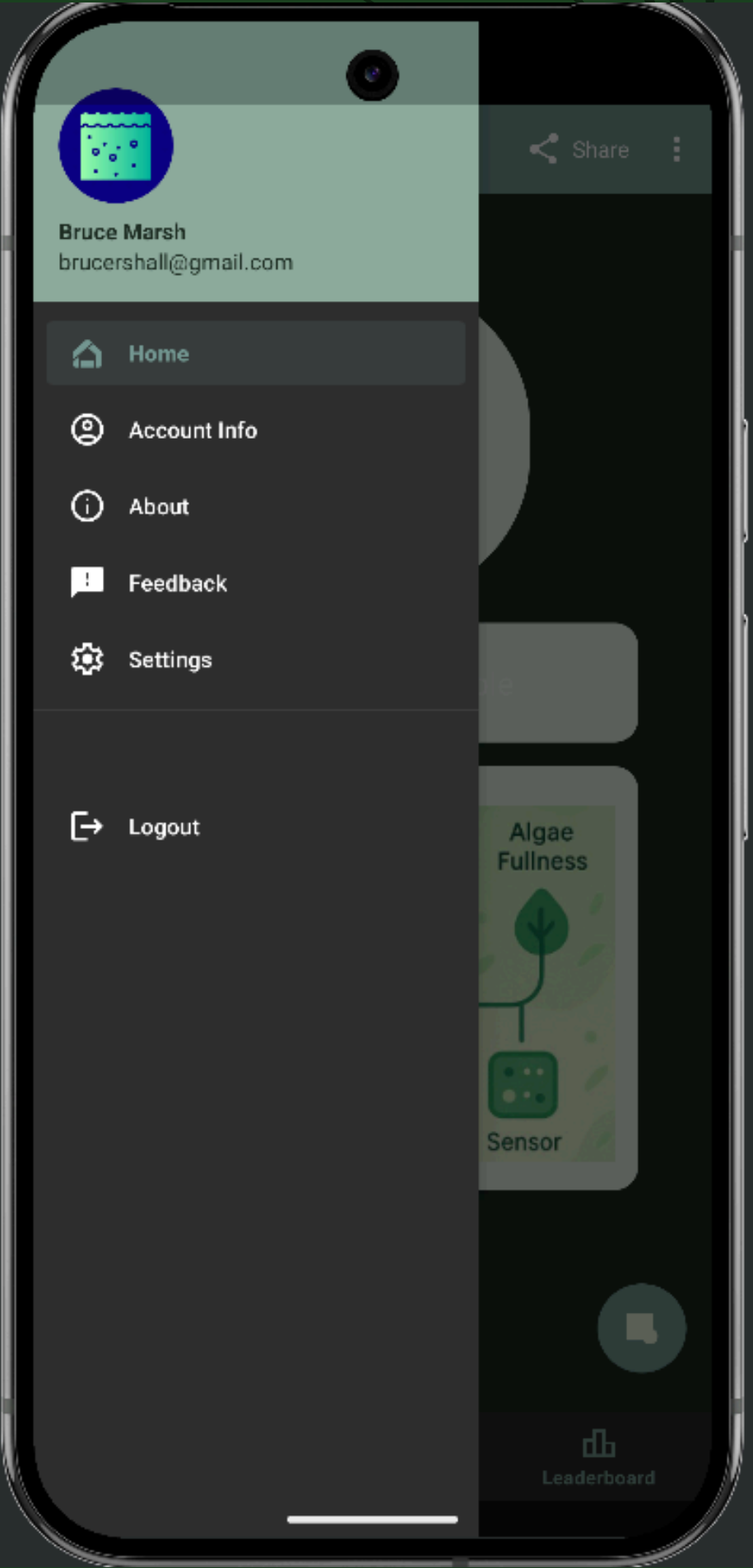
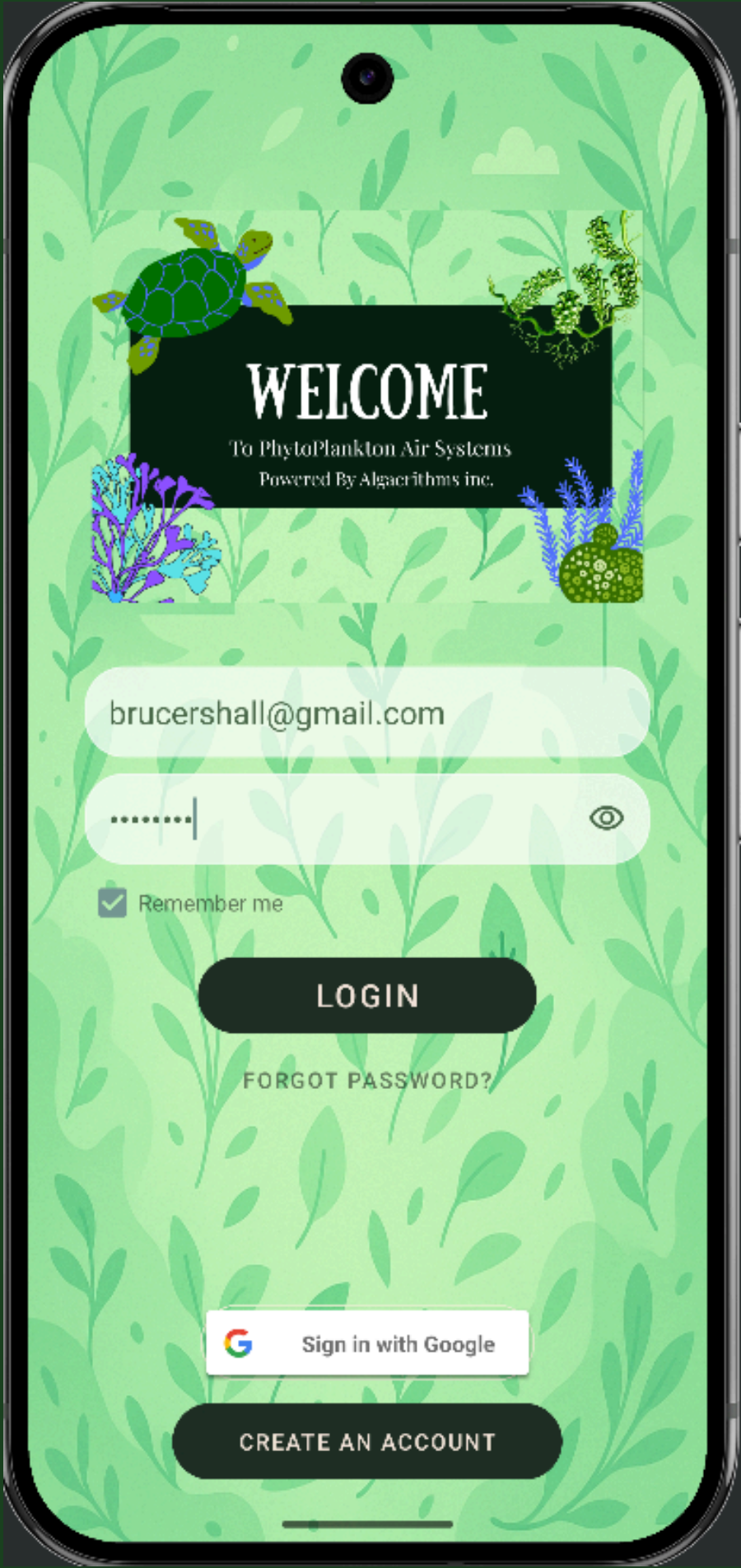
- Android application built to display all live sensor readings sent from the Raspberry Pi.
- Secure Firebase Authentication allows users to log in with stored credentials.
- App retrieves real-time values from the Realtime Database (e.g., algaeHealth, light, turbidity, proximity, CO₂ converted, timestamp).
- Dashboard Screen shows current environmental values in a clean, user-friendly layout.
- Leaderboard – compares users based on captured CO₂ metrics
- Insights – displays weekly/monthly performance trends
- Provides an educational and interactive experience by visualizing environmental changes instantly.

Demonstration Setup — 3D Printed Enclosure & Hardware Integration

- System housed in a custom 3D-printed enclosure holding the Raspberry Pi 4, PCB, wiring, and all sensors.
- Enclosure includes individual compartments for sensors to improve airflow and stable placement.
- 3D-printed design provides clean cable management and protects components during demos.
- Sensors connect via Qwiic connectors for quick swap, maintenance, and reliability.
- Custom PCB replaces the breadboard for reduced wiring errors and better stability.
- Raspberry Pi streams real-time data (air quality, light, proximity, turbidity) to Firebase, displayed instantly in the mobile app.
- Compact setup demonstrates smooth hardware integration and live data flow end-to-end.

DEMO

Glimpse Of the App



An underwater photograph showing a large school of small, silvery fish swimming in clear blue water. In the foreground, there are large, yellowish-green seaweed leaves. The image is partially obscured by a dark, irregular shape on the right side, which serves as a background for the text.

Why Our Project Breaks the Mold

Our project goes beyond typical app development — we integrated a real algae system to bring biology and tech together. We built for impact, not just function, adding a climate education screen, gamified tracking, and real-world service features. Every detail, from reusable algae pods to user-focused design, was made to create a product that's sustainable, engaging, and built to last.



CENG322 Software
Project

Thank you

